

Industrial Digital Twin Input Layer

Architecture for Horizon and Zephyr Hot Rolling Mills

Part 1: Process Flow Analysis

The implementation of a high-fidelity industrial digital twin requires a comprehensive understanding of the physical manufacturing processes it seeks to simulate. The Horizon and Zephyr facilities both operate within the domain of steel manufacturing, specifically functioning as hot rolling mills designed to transform solidified steel into coiled strips. However, the architectural topologies of these two plants diverge slightly to accommodate different market demands and throughput capacities, necessitating distinct digital twin process mapping. The Horizon factory operates as a conventional Hot Strip Mill (HSM). The raw materials consist of thick, continuously cast steel slabs, typically ranging from 200 mm to 250 mm in thickness, which are transported from a slab yard. The material flow at Horizon begins with the Slab Reheating Furnace, an intermediate stage where these massive slabs are thermally soaked to austenitizing temperatures.¹ Following thermal equalization, the material passes through a High-Pressure Descaler to violently shear off the primary iron oxide scale.³ The clean slab then enters the Roughing Mill—a massive reversing stand or series of continuous stands—which executes primary thickness reduction, elongating the slab into a transfer bar. This transfer bar is subsequently fed into a Tandem Finishing Mill consisting of five to seven closely coupled roll stands equipped with Hydraulic Automatic Gauge Control (HAGC) to achieve the final micro-millimeter thickness tolerance.⁵ The incandescent steel strip is then transported across a Laminar Cooling Bed, where precise water jets orchestrate the metallurgical phase transformation before the strip reaches the Down Coiler, which winds the material into the final output product: Hot Rolled Coils (HRC).⁷

Conversely, the Zephyr factory utilizes a Thin Slab Casting and Rolling (TSCR) or Compact Strip Production (CSP) architecture. The raw material here is liquid steel directly cast into thin slabs (approximately 50 mm to 90 mm thick). This continuous flow bypasses the traditional ambient-temperature slab yard entirely.⁹ The intermediate stages begin with a tunnel equalization furnace, functioning primarily to homogenize the temperature of the already-hot thin slab rather than reheating it from an ambient state. The material flow proceeds through a descaler and directly into the finishing mill stands, omitting the massive roughing mill phase entirely.⁹ This dynamic liquid core reduction and lean organizational structure drastically reduce capital floor space and thermal energy requirements, but present unique digital twin challenges regarding the tight coupling between the continuous caster and the rolling stands.⁹ The output product remains the Hot Rolled Coil, though often tailored for specialized downstream applications.

The dependencies between equipment in both factories dictate a strict, unyielding sequence. A temporal or physical anomaly in any upstream node instantaneously cascades to the next. The

digital twin must map these dependencies accurately. For Horizon, the dependency diagram follows: Slab Yard → Reheating Furnace → Primary Descaler → Roughing Mill → Secondary Descaler → Tandem Finishing Mill → Run-Out Table Cooling → Down Coiler. For Zephyr, the dependency maps as: Liquid Steel Caster → Thin Slab Tunnel Furnace → Descaler → Tandem Finishing Mill → Run-Out Table Cooling → Down Coiler.

Part 2: Registry Table Verification

The foundational step in simulating an industrial Internet of Things (IIoT) environment is verifying the accuracy of the equipment registry tables. Existing enterprise resource planning (ERP) registries often contain generalized, outdated, or incomplete engineering fields that fail to capture the thermodynamic and kinematic realities required for a predictive machine learning model. The verified and corrected technical baseline for both the Horizon and Zephyr facilities is detailed below.

Equip ment Name	Prima ry Task	Input Mater ial	Outp ut Mater ial	IoT Senso rs Used	Senso r Data Types	Opera ting Envel ope	Failur e Physi cs / Malfu nction Mech anism s	ISO / Indust ry Stand ards
Slab Rehea ting Furna ce	Auste nitize steel slabs via therm al radiati on	Cast Steel Slabs (Ambi ent/W arm)	Heate d Slabs (~120 0°C)	Therm ocoup les, Pyrom eters, Flowm eters	Tempe rature (°C), Gas Flow (m³/h)	600–1 200°C , Contin uous walkin g beam	Refrac tory degra dation , Burner misfire , Walkin g beam hydra ulic leak	ISO 14001, ISO 9001, ISO 55000 ⁷

High-Pressure Descaler	Remove iron oxide scale	Heated Slabs with Surface Scale	Descal ed Slabs	Pressure Transducers, Flowmeters	Pressure (Bar), Flow Rate (L/min)	380–400 Bar fluid pressure ⁴	Pump cavitation, Nozzle erosion, Valve fatigue	ISO 14224, IEC 61508
Roughing Mill (Horizon Only)	Primary thickness reduction	Descal ed Slabs	Transfer Bar	Load Cells, Encoders, Accelerometers	Force (kN), Torque (Nm), Vibration (mm/s ²)	5,000–12,000 kW Motor Power ⁸	Gear wear, Roll fatigue, Bearing spallation	ISO 10816, ISO 14224
Tandem Finishing Mill (HAGC)	Final gauge precision control	Transfer Bar / Thin Slab	Thin Steel Strip	LVDT, Servo-pressure sensors	Position (μm), Pressure (Bar)	±0.05 mm tolerance, 1500 m/min speed ⁸	Servo valve hysteresis, Cylinder seal leakage, Mill chatter	ISO 4406, ASTM A568
Laminar Cooling Bed	Microstructural phase transformation	Austenitic Steel Strip	Ferritic/Bainitic Strip	Pyrometers, Flowmeters, Valve actuators	Temperature (°C), Flow Rate (L/min)	800°C down to 550°C at 10–50 °C/sec	Header blockage, Uneven cooling flow, Valve stiction	ISO 9001, ASTM E112

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Down Coiler	Wind hot strip into transportable coils	Cooled Steel Strip	Hot Rolled Coil (HRC)	Tension meters, Speed encoders	Tension (MPa), RPM	Strip tension variable by grade	Mandrel collapse failure, Wrapper roll misalignment	ISO 9001, OSHA 1910.212

Legacy registry tables previously listed the high-pressure descender operating envelope generically as "high pressure," which is mathematically insufficient for physical simulation. This has been corrected to the specific 380–400 bar requirement necessary for effective oxide removal without inducing thermal shock.⁴ Furthermore, legacy entries for the Finishing Mill often cited generic "position sensors," which have been explicitly updated to Linear Variable Differential Transformers (LVDT), as these are the exact transducers utilized in Hydraulic Automatic Gauge Control (HAGC) cylinders to measure micro-millimeter displacement.⁶ The inclusion of specific failure physics, such as "Pump Cavitation" and "Servovalve Hysteresis," provides the causal targets that the downstream machine learning models must be trained to detect.

Part 3: Equipment-Level Deep Research

To parameterize the local machine learning models and the simulated digital twin, exhaustive research into the operational constraints, standard operating procedures, and maintenance frameworks of each primary asset is required.

Slab Reheating Furnace

The walking-beam slab reheating furnace represents the primary thermal bottleneck in the hot rolling process. It utilizes radiant heat transfer from combusted mixed gases to elevate the core temperature of a steel slab to approximately 1200°C, rendering it malleable for plastic deformation.² The capacity of a typical modern furnace ranges from 250 to 400 tons per hour. The operating parameters dictate a strict thermal gradient; the surface-to-core temperature differential must be minimized to prevent asymmetric rolling forces downstream.¹⁴ The primary design constraint is the refractory lining, which serves as thermal insulation and is highly susceptible to thermal shock.¹⁰

Standard Operating Procedures (SOPs) for the furnace demand a rigorous startup sequence. The atmosphere must initially be purged with inert nitrogen to prevent explosive atmospheric mixtures, followed by the sequential ignition of pilot burners. Normal operation requires the continuous synchronization of the walking beam pace with the mill entry speed, ensuring that

slabs are discharged precisely when the roughing mill is clear, thereby eliminating inter-slab delays.¹⁰ Shutdown procedures dictate a controlled thermal descent, reducing fuel gas flow over a 24-hour period while maintaining combustion air circulation to cool the refractory bricks evenly. Preventive maintenance standards require daily visual inspections of the water-cooling circuits protecting the walking beam skids. Weekly checks involve calibrating the multi-zone thermocouples against master optical pyrometers. Predictive maintenance practices focus heavily on refractory lining campaign management, utilizing external thermal imaging to calculate remaining campaign life and prevent structural breakthroughs.¹⁰ The equipment operates under strict adherence to ISO 14001 for emissions and environmental management.¹⁰

High-Pressure Descaler

The High-Pressure Descaler acts as the primary quality control gate. As steel heats, oxygen reacts with the surface to form a thick, brittle layer of iron oxide (scale). If rolled into the substrate, this scale creates irreversible surface defects. The descender employs multi-stage centrifugal pumps feeding a spray header equipped with tungsten carbide nozzles to blast the slab with water at 380 to 400 bar.⁴ Typical flow rates reach 5,000 liters per minute. Design constraints include the impingement angle of the jets, which must be perfectly calibrated to shear the scale without causing excessive thermal quenching of the steel slab. Startup SOPs necessitate priming the massive centrifugal pumps and opening bypass valves to establish a stable hydrodynamic flow before the high-pressure headers are brought online. Normal operation relies on optical sensors detecting the approaching slab to trigger the accumulator valves, firing the jets precisely as the slab enters the descaling box. Shutdown sequences require a full pressure bleed-off before halting the main pump motors to mitigate severe water hammer effects that could rupture the high-pressure piping. Preventive maintenance involves a strict daily inspection of the accumulator pressure bladders. Weekly maintenance focuses on inspecting the tungsten carbide nozzles for localized erosion, which alters the spray pattern and decreases impact force.¹¹ Predictive maintenance utilizes high-frequency acoustic emission sensors on the pump casings to detect the onset of fluid cavitation. Applicable standards include API 610 for centrifugal pumps and IEC 61508 for the automated safety interlocking valves.

Tandem Finishing Mill (with HAGC)

The Tandem Finishing Mill is the kinematic heart of the factory, responsible for the final geometric shaping of the steel strip. It consists of five to seven successive stands, where the steel is pulled under immense tension and compressed by massive work rolls. The critical sub-system here is the Hydraulic Automatic Gauge Control (HAGC), which utilizes four-way servovalve-controlled hydraulic cylinders to adjust the roll gap in milliseconds.⁶ The mill operates under extreme loads ranging from 10 to 20 MN of rolling force, reducing strip thickness down to tolerances of ± 0.05 mm while traveling at speeds up to 1500 meters per minute.⁸

Startup procedures involve a calibration process known as "mill spring compensation," where the rolls are brought into direct contact (kissing) under high hydraulic pressure to zero the LVDT

sensors and measure the elastic deformation of the mill housing.¹⁵ Normal operation requires the complex coordination of inter-stand tension; if tension is too high, the strip will tear, but if it is too low, the strip will buckle and cause a catastrophic "cobble." Shutdown protocols involve relieving all hydraulic pressure and separating the work rolls to prevent localized thermal expansion damage from radiant heat. Preventive maintenance demands daily fluid sampling to ensure the hydraulic oil meets ISO 4406 cleanliness standards, as particulate contamination will score the delicate servovalve spools.¹¹ Predictive maintenance relies heavily on vibration analysis of the massive backup and work roll bearings, mapping the frequency domain to identify outer race or inner race spallation before catastrophic failure.¹¹

Part 4: Factory-Level SOP and Maintenance Standards

The transition from reactive firefighting to proactive, digital twin-driven predictive maintenance requires a foundational shift in factory-wide standard operating procedures. As publicly available documentation for the specific internal operations of Horizon and Zephyr is inherently proprietary, industry-standard equivalents have been synthesized based on ISO 9001 (Quality Management), ISO 55000 (Asset Management), ISO 14224 (Reliability and Maintenance Data), and ISO 17359 (Condition Monitoring).¹⁰

The factory-wide SOP establishes a rigid framework for production orchestration. Both the Horizon and Zephyr pipelines operate on a synchronized push-pull logic dictated by the downstream bottleneck. For instance, the Slab Reheating Furnace acts as the primary thermal pacemaker. Inter-facility buffers—such as the space between the roughing mill and finishing mill—must be strictly monitored; if the buffer exceeds 80% capacity, upstream pushing must halt to prevent irreversible thermal energy loss to the ambient environment. This requires an integrated Center of Excellence (CoE) dashboard that provides a macroscopic view of material flow.¹⁷

The inspection schedules are stratified across three distinct tiers. Tier 1 inspections are automated, continuous, and driven by the IIoT sensor layer, recording data at millisecond intervals to populate the digital twin. Tier 2 inspections are executed by shift technicians on a daily and weekly basis, focusing on physical phenomena undetectable by sensors, such as fluid leakages, abnormal localized odors, or visual degradation of safety guarding. Tier 3 inspections encompass comprehensive monthly and quarterly non-destructive testing (NDT), including ultrasonic testing of mill housings and thermographic surveys of electrical switchgear. The maintenance hierarchy is explicitly structured to prioritize condition-based interventions over reactive fixes. Run-to-failure (reactive maintenance) is strictly prohibited for any asset residing on the critical path of the main product flow. Preventive maintenance—relying on static, time-based component replacement—is utilized only for components with highly predictable wear rates, such as fluid filters or contact wipers. The digital twin facilitates the highest tier: Predictive Maintenance (PdM). In this regime, maintenance interventions are dynamically scheduled solely based on machine learning models forecasting the Remaining Useful Life (RUL) of the component, thereby maximizing asset utilization while virtually eliminating unplanned downtime.⁵

Alarm handling procedures dictate the human-in-the-loop response to anomalies detected by

the edge AI models. A Level 1 (Advisory) alarm indicates a statistical deviation from the norm, such as a slow drift in bearing temperature. The operator acknowledges the alarm, and the system automatically generates a predictive maintenance ticket for the next scheduled downtime window. A Level 2 (Warning) alarm signifies that a degradation threshold has been breached, triggering an immediate on-the-ground visual inspection by the shift technician to verify the AI's diagnosis. A Level 3 (Critical) alarm indicates imminent catastrophic failure; this triggers the safety programmable logic controllers (PLCs) compliant with IEC 61508 to initiate an automated interlocking sequence, safely halting the affected equipment line without human intervention.

Shutdown and emergency procedures are designed to safely arrest thousands of tons of moving, incandescent steel. In the event of a catastrophic loss of utility (e.g., total power failure or loss of cooling water), gravity-fed overhead emergency tanks deploy cooling water to prevent the mill rolls from cracking under extreme thermal stress. Simultaneously, dynamic braking resistors absorb the regenerative energy from the main DC drives, halting the spinning mass of the tandem mill within seconds. Slabs currently transiting the mill are mechanically pushed off the main line to secure holding areas, preventing them from fusing to the rollers as they cool.

Part 5: Sensor and IoT Research

The efficacy of the Industrial Digital Twin is entirely dependent on the fidelity, resolution, and accuracy of the Input Layer. The sensor telemetry must capture the subtle thermodynamic and kinetic deviations that precede mechanical failure. The digital twin architecture ingests 49 distinct process parameters (denoted X_1 through X_{49}) that form the multivariate feature space for detecting defects such as the Alpha anomaly.¹⁷

Sensor Type	Target Equipment / Application	Engineering Units	Sampling Rate	Typical Operating Range	Normal Values	Alarm Limit	Failure Threshold
Optical Pyrometer	Reheating Furnace Exit, Run-Out Table	°C	10 Hz	600–1250°C	1185–1215°C	< 1170°C	< 1150°C
Piezoelectric Acceler	Tandem Mill Backup & Work	mm/s ² (RMS) & g-force	10 kHz to 50 kHz	1.0–15.0 mm/s ²	1.5–4.5 mm/s ²	> 7.1 mm/s ²	> 11.0 mm/s ²

ometer	Roll Bearing s	s					
Piezoresistive Pressure Transducer	High-Pressure Descaler Header	Bar	100 Hz	0–450 Bar	380–400 Bar	< 350 Bar	< 320 Bar
LVDT (Position Encoder)	Hydraulic AGC Cylinders	μm	1 kHz	0–5000 μm	Target ± 20 μm	± 50 μm	± 100 μm
Current Transformer (Hall Effect)	Main Mill Drive Motors	Amps (A)	1 kHz	0–10,000 A	3000–5000 A	> 5500 A	> 6500 A
Acoustic Emission Sensor	Main Drive Gearboxes	dB / kHz	100 kHz	0–120 dB	Baseline	Baseline + 10dB	Baseline + 20dB
Inline X-Ray / Isotope Thickness Gauge	Finishing Mill Exit	mm	100 Hz	0.5–25.0 mm	1.50–5.00 mm	± 0.05 mm	± 0.10 mm
Load Cell (Strain	Roughing / Finishing	kN / MN	500 Hz	0–30 MN	10–20 MN	> 22 MN	> 25 MN

Gauge)	g Mill Housing s						
Laser Doppler Velocimeter	Strip Speed over Cooling Bed	m/min	100 Hz	0–2000 m/min	1000–1500 m/min	± 5% variation	± 10% variation
Tensio meter	Down Coiler Mandrel	MPa	200 Hz	0–100 MPa	Grade dependent	± 10% tension	Loss of tension

The sampling rates specified in this matrix are critical for signal processing accuracy. High-frequency phenomena, such as the initial micro-cracking of a bearing inner race or the high-frequency chatter in a tandem mill, require accelerometers sampling at 10 kHz or greater to satisfy the Nyquist-Shannon sampling theorem and capture the relevant harmonics without aliasing.¹⁸ Conversely, thermodynamic parameters measured by pyrometers change at a much slower rate due to the immense thermal inertia of steel, allowing for a 10 Hz sampling rate without losing process fidelity. The LVDTs and load cells require 1 kHz sampling to properly map the rapid elastic deformation of the mill housing and the millisecond response of the servovalves to incoming strip thickness variations.⁶

Part 6: Failure Physics

Machine learning models require robust feature engineering based on the actual physical degradation pathways of the machinery. The digital twin must correlate statistical anomalies with deterministic physical mechanisms to provide engineers with interpretable Root Cause Analyses.¹⁷

Bearing Wear and Spallation

The backup and work rolls of the tandem mill are subjected to immense cyclic loading, often exceeding 20 MN. Over millions of revolutions, subsurface shear stresses initiate micro-cracks below the bearing raceway. As these cracks propagate to the surface, microscopic flakes of metal break away—a process known as spallation. The early indicators are microscopic metallic debris in the lubrication oil and extremely high-frequency, low-amplitude acoustic emissions. The sensor signature transitions to distinct vibration peaks at the bearing's specific defect frequencies (e.g., Ball Pass Frequency Outer race - BPFO). The root causes include prolonged heavy loading, inadequate lubrication film thickness, or improper installation. The progression pattern begins linearly but turns aggressively exponential once spallation creates macroscopic

pits on the raceway, ultimately leading to catastrophic seizure.¹¹

Cavitation in High-Pressure Descaler Pumps

Cavitation occurs in the centrifugal pumps of the descaler when the localized fluid pressure at the impeller inlet drops below the vapor pressure of the water. This causes the water to briefly boil, forming vapor cavities. As these bubbles are swept into the high-pressure volute of the pump, they violently implode, generating micro-jets of water that strike the impeller blades with forces exceeding the yield strength of the metal. The physical mechanism erodes the metal surface, creating a pitted, sponge-like texture. Early indicators include a slight drop in volumetric flow and a distinct, high-frequency "crackling" noise. Sensor signatures present as broadband noise in the high-frequency acoustic emission spectrum (20–50 kHz), eventually manifesting as sub-synchronous vibration peaks.¹¹ The root causes are typically inadequate Net Positive Suction Head available (NPSHa), clogged inlet filters, or operating the pump too far right on its performance curve. The progression is self-accelerating; initial pitting disrupts hydrodynamic flow, exacerbating pressure drops and accelerating further cavitation.

Hydraulic Servovalve Hysteresis and Seal Leakage

The Hydraulic Automatic Gauge Control (HAGC) relies on ultra-precise servovalves to meter fluid to the main cylinders. Contamination in the hydraulic fluid—often microscopic metallic wear particles or degraded seal elastomers—acts as a lapping compound, gradually rounding the sharp metering edges of the valve spool. Simultaneously, dynamic friction between the cylinder piston and the seals causes elastomeric wear.¹⁵ The physical mechanism results in internal fluid bypass. Early indicators are sluggish step-response times to control commands and minor position overshoots. Sensor signatures reveal a growing phase lag between the command signal and the LVDT position feedback, alongside a requirement for continuously higher pump pressure to maintain the same holding force. The root cause is universally related to inadequate fluid filtration failing to maintain ISO 4406 cleanliness standards.¹¹ The progression is slow and linear, but directly impacts the final product tolerance, leading to gauge variations.

Thermal Fatigue

The walking beams inside the reheating furnace endure severe cyclical thermal loading. They are exposed to 1200°C temperatures while supporting the cold, incoming slabs, followed by relative cooling when unloaded. This differential thermal expansion and contraction generates severe cyclical stress, causing micro-cracking in the refractory insulation and the underlying metallic skids. The physical mechanism is identical to low-cycle thermal fatigue. Early indicators include localized cold spots on the exiting slabs, caused by heat being conducted away into the damaged skid structure. Pyrometers detect inconsistent temperature profiles across the width of the slab. The root causes include operating the furnace with excessively rapid heating or cooling ramps during startup and shutdown, or loss of internal water cooling to the skids. The progression pattern is insidious; it remains hidden beneath the refractory layer until full structural failure occurs, leading to a collapsed walking beam.

Chatter (Excess Vibration) in Tandem Mills

Mill chatter is a complex, self-excited vibrational phenomenon that occurs in the finishing mill. It arises when the natural resonant frequency of the mill housing couples with the rolling process mechanics—specifically variations in strip tension, roll friction, or hydraulic cylinder dynamics.¹⁸ The physical mechanism involves a regenerative feedback loop where a slight variation in strip thickness causes a variation in rolling force, which physically bounces the massive mill rolls, imprinting a wave pattern onto the steel strip. Early indicators are high-pitched whining or rumbling noises emitting from the stand, and subtle, periodic gauge variations in the final strip. Sensor signatures present as massive, dominant frequency spikes in the vibration spectrum (typically around 100–200 Hz for third-octave chatter). The root causes include running the mill at critical resonant speeds, inadequate rolling lubrication, or worn backup roll bearings. The progression is instantaneous and violent; chatter can rapidly escalate in seconds, tearing the steel strip or shattering the work rolls.

Alpha Defect Formation

The "Alpha defect" represents a localized metallurgical anomaly that must be predicted by the $X1-X49$ variable matrix.¹⁷ The physical mechanism typically involves a hard, highly dense inclusion—often enriched in nitrogen or complex oxides—that fails to deform plastically at the same rate as the surrounding austenitic steel matrix during the high-compression rolling phases.²¹ Because this node resists deformation, it creates immense localized stress concentrations. Early indicators include abnormal thermal gradients across the slab surface and subtle spikes in the localized rolling force as the inclusion passes through the roll bite. The sensor signatures manifest as sudden, microsecond-duration spikes in the load cells of the roughing and finishing stands, coupled with distinct, high-frequency vibration transients.²¹ The root causes trace back to variations in the incoming continuous casting chemistry, improper tundish flux practices, or inadequate thermodynamic soaking time in the reheating furnace.

Part 7: ML Dataset Research

The successful training of localized edge AI models requires access to high-quality, annotated datasets that capture a wide variance of operational states, process parameters, and diverse fault classes. The research has identified several crucial public and proprietary datasets suitable for time-series forecasting, anomaly detection, and predictive maintenance in hot rolling mill environments.

Dataset Name	Source Link	Sensor Variables	Sampling Frequency	Fault Classes / Labels	Number of Records	Equipment Type
Horizon Defect	Project ATAL	X1 through	Stage-based	Binary target Y	1,352 Training,	Hot Rolling

Dataset (Internal Upload)	Repository (train.csv)	X49	(Aggregated)	(Alpha defect occurrence: 1 or 0)	339 Testing	Mill Pipeline ¹⁷
Benchmark Datasets for Predictive Maintenance in Steel	Zenodo (Record 11469702) ²²	Rolling force, torque, speed, tension, gap, thickness reduction, motor power	100 Hz	Variable anomalies related to cold rolling failures	Six unique simulated datasets	5-Stand Tandem Mill ⁵
Industrial Milling Tool Life Dataset	Kaggle ²⁵	Spindle Speed, Feed Rate, Cutting Depth, Vibration X/Y, Acoustic, Spindle Load	1 kHz	Tool wear progression, RUL estimation	10,000+	Rotational Milling Equipment ²⁵
Automated Fatigue Life Prediction of Welded Steel	Kaggle ²⁶	Stress, Strain, Geometry, NDT optical imaging	Variable	Fatigue behavior, crack propagation	5,000+	Steel Structures / Welding ²⁶
MIMI (Malfunctioning Industrial)	Zenodo / IEEE Dataport	High-frequency acoustic audio	16 kHz	Normal vs. Anomalous audio	24,000+ audio clips	Valves, Pumps, Sliders

I Machine Investiga tion)		streams		signature s		
NASA IMS Bearing Dataset	NASA Prognosti cs Center of Excellenc e ²⁷	High-freq uency Vibration (Accelerometers)	20 kHz	Inner race, outer race, and rolling element defects	100M+ data points	Industrial Bearings

The internal Horizon dataset provides the exact 49-variable feature space required to isolate the Alpha defect, serving as the primary classification benchmark where a 100% recall metric must be achieved.¹⁷ However, because actual failure data is rare, the Jakubowski synthetic dataset provides a mathematically rigorous foundation for pre-training models on rolling mill kinematics. While modeled on a cold mill, the physics of strip tension, roll gap, and motor torque mathematically parallel the finishing stands of the Horizon and Zephyr hot mills, making it an ideal candidate for transfer learning.⁵ The NASA IMS dataset is utilized exclusively for training the localized anomaly detection models tasked with monitoring the heavy backup roll bearings, providing the high-frequency run-to-failure signatures required to calculate Remaining Useful Life (RUL).²²

Part 8: Equipment-Specific ML Models

Meeting the strict architectural constraints—zero false negatives for defect detection and less than 10% false positive alarm rates—demands a hierarchical ensemble of machine learning models tailored to the specific physics of each equipment type.¹⁷

Traditional Models (Random Forest & XGBoost)

Traditional ensemble tree-based models excel at handling structured, tabular data where complex, non-linear relationships exist between discrete process parameters.

- **Suitability:** These models are the optimal choice for interpreting the anonymized, multi-stage X1-X49 dataset provided in train.csv to predict the Alpha defect.¹⁷
- **Inputs / Outputs:** Inputs comprise the 49 static or aggregated process parameters representing a single steel coil. The output is a binary probability score indicating the likelihood of defect formation.
- **Training Data Requirements:** Requires a historically labeled dataset with a sufficient balance of defect (Y=1) and non-defect (Y=0) instances.
- **Computational Complexity:** Low. These models are highly interpretable, allowing engineers to extract feature importance scores to determine exactly which thermodynamic or kinematic stage is inducing the defect.

Sequence Models (Bi-LSTM & GRU)

Recurrent Neural Networks, specifically Bidirectional Long Short-Term Memory (Bi-LSTM) networks, are designed to process sequential data, making them ideal for capturing degradation trends over time.

- **Suitability:** Highly effective for predicting the Remaining Useful Life (RUL) of continuously degrading components, such as the descaler pump impellers experiencing cavitation or the HAGC servovalves suffering from hysteresis.
- **Inputs / Outputs:** Inputs are time-series sequences of vibration, acoustic, and pressure sensor data. The output is a continuous regression curve estimating the remaining operational hours before the failure threshold is breached.
- **Training Data Requirements:** Requires run-to-failure datasets (such as the NASA IMS dataset) to map the full lifecycle of a component's degradation.
- **Computational Complexity:** Moderate. Inference is rapid, allowing these models to be deployed directly on edge-compute nodes near the mill stands for real-time monitoring.

Transformer Models (PatchTST & Time Series Transformer)

Transformers utilize self-attention mechanisms to weigh the significance of different parts of the input data, regardless of their temporal distance.

- **Suitability:** Unparalleled at capturing long-term dependencies in high-frequency, multivariate data, such as the 10 kHz acoustic and vibration streams emitted from the tandem mill gearboxes and roll stands.
- **Inputs / Outputs:** Inputs are patched, overlapping windows of multi-variate sensor streams. Outputs are forecasted future sensor values; anomalies are flagged when the actual incoming data sharply deviates from the Transformer's highly accurate forecast.
- **Training Data Requirements:** Extremely large volumes of continuous time-series data.
- **Computational Complexity:** High. Training requires massive parallelization on cloud or centralized high-performance computing clusters (such as the iROC CoE), though quantized inference can be executed locally.¹⁷

Autoencoder Models (Variational Autoencoder - VAE)

Autoencoders compress input data into a lower-dimensional latent space and then attempt to reconstruct the original input.

- **Suitability:** Perfect for unsupervised anomaly detection in industrial environments where historical breakdown logs are sparse or non-existent, a common scenario in tightly controlled plants like Tata Steel.
- **Inputs / Outputs:** Inputs are multi-sensor arrays during verified normal operation. The output is a reconstruction error. If a novel mechanical fault occurs, the VAE fails to reconstruct the signal accurately, and the surging reconstruction error triggers an alarm.
- **Training Data Requirements:** Requires solely "healthy" operational data, completely bypassing the need for labeled fault data.
- **Computational Complexity:** Moderate.

Graph Models (Graph Neural Networks - GNN)

Graph models process data structured as nodes and edges, allowing them to map the physical topology of the manufacturing floor.

- **Suitability:** Designed to model cross-equipment dependencies. Nodes represent equipment (Furnace, Descaler, Mill), and edges represent the physical material flow and thermodynamic transfer.²⁸
- **Inputs / Outputs:** Inputs are spatial-temporal graphs populated with sensor data at each node. Outputs are node-level classifications that trace the exact root cause of an anomaly propagating through the multi-machine setup.
- **Training Data Requirements:** Requires synchronized, time-stamped data across the entire factory pipeline to establish causal links.
- **Computational Complexity:** High, due to the message-passing algorithms across large industrial networks.

Foundation Models (TimesFM, Chronos, Lag-Llama)

These represent the bleeding edge of zero-shot time-series forecasting, pre-trained on billions of data points across diverse domains.

- **Suitability:** Highly suitable for rapid deployment in new factory sections where historical data has not yet been collected. They act as out-of-the-box forecasters for general parameters like motor temperature or fluid pressure.
- **Inputs / Outputs:** Raw, univariate or multivariate time series. Outputs probabilistic forecasts with confidence intervals.
- **Training Data Requirements:** None required for initial deployment (zero-shot capability), though fine-tuning improves accuracy.
- **Computational Complexity:** Extremely high for training, moderate for inference.

Part 9: Cross-Equipment Dependency Dataset

A defining characteristic of the steel manufacturing pipeline is that localized machine faults do not exist in isolation; they create time-lagged, cascading failures downstream.¹⁷ The digital twin cannot treat equipment as isolated nodes; it must mathematically represent the entire sequence as a coupled ecosystem.²⁹

Consider the cascading failure paradigm originating in the Slab Reheating Furnace. If a specific heating zone experiences a minor burner failure or a drop in mixed-gas pressure, a slab may exit the furnace at 1160°C instead of the optimal 1200°C. As this slab traverses to the High-Pressure Descaler, the slightly cooler, more tenacious iron oxide scale resists hydraulic shearing, leaving microscopic scale patches on the steel surface. As this compromised slab enters the Roughing Mill, the cooler steel matrix exhibits a substantially higher yield strength. This triggers a massive, instantaneous spike in the required rolling force and motor torque, which physically deflects the mill housing and accelerates the degradation of the work roll bearings. Finally, as the slab enters the Tandem Finishing Mill, these upstream inconsistencies manifest violently as mill chatter and localized stress concentrations, ultimately forging the

critical Alpha defect into the final Hot Rolled Coil.¹⁷

To capture these complex phenomena, the Digital Twin must construct a multivariate causal dataset. Traditional correlation matrices fail in this environment because they cannot distinguish between the root cause and the downstream symptom. Instead, methodologies such as the Nonlinear Multivariate Lasso Granger (NMLG) algorithm must be employed.³⁰

The construction of these combined datasets requires an architecture that integrates temporal event propagation. First, all IoT sensor data must be stamped with synchronized, high-precision timestamps (e.g., using Precision Time Protocol, IEEE 1588) to align the 10 kHz vibration data of the mill with the 10 Hz thermodynamic data of the furnace. Second, a many-to-one neural network architecture equipped with input attention mechanisms adaptively identifies the driving variables of a fault.³⁰ By running Monte Carlo-based significance tests on these variables, the system eliminates ambiguous or spurious causal pathways, creating a deterministic map.³⁰ This allows the Agentic AI to look at a gauge variation in the finishing mill and correctly trace the root cause backward in time to a 40°C temperature drop that occurred two hours prior in the reheating furnace.

Part 10: Simulation Data Generation

Because historical failure data containing catastrophic breakdowns is exceedingly rare in tightly controlled environments like Horizon and Zephyr, the reliance on purely data-driven ML models is perilous. Synthetic data generation via high-fidelity, physics-based simulation is mandatory for pre-training the ML models and validating the digital twin's anomaly detection capabilities prior to physical deployment.⁵

Physics-Based Simulators

- **MATLAB Simulink & Simscape:** Ideal for modeling the complex mechanical-hydraulic coupling of the HAGC system. The dynamics involve the fluid bulk modulus, cylinder compliance, and servovalve fluid mechanics.¹²
- **OpenModelica:** Highly suited for thermodynamic modeling of the Slab Reheating Furnace, utilizing equation-based, object-oriented modeling to solve multi-zonal radiant heat transfer and fluid flow.
- **AnyLogic:** Utilized for discrete-event simulation to model the macroscopic material flow and logistics across the factory floor, pinpointing queuing bottlenecks between the descaler and the roughing mill.

Governing Equations and State Variables

To guarantee the empirical fidelity of the generated synthetic datasets, the simulators must solve the foundational physics equations governing the processes:

1. Descaler Fluid Dynamics

The high-pressure water jets must be modeled using the Navier-Stokes equations for incompressible flow. The continuity and momentum equations dictate the fluid velocity and pressure gradients:

$$\nabla \cdot (\rho \mathbf{v}) = 0$$

$$\nabla \cdot (\rho \mathbf{v} \mathbf{v}) = -\nabla p + \nabla \cdot (\boldsymbol{\tau}) + \rho \mathbf{g} + \mathbf{F}$$

where p is static pressure, ρ is density, $\boldsymbol{\tau}$ is the stress tensor, $\mathbf{v}$ represents fluid velocity, and $\mathbf{F}$ is the external body force.¹³

- **State Variables:** Fluid pressure, fluid velocity, volumetric flow rate.
- **Sensor Variables:** Simulated pressure transducer outputs, acoustic emission spectrum.
- **Fault Injection:** Impeller degradation is simulated by incrementally altering the localized pressure p field at the inlet boundary condition to artificially induce the thermodynamic state of cavitation.

2. Hydraulic AGC System Dynamics

The HAGC system acts as a mechanical-electric-hydraulic coupled field. The dynamic model for the displacement sensor and the actuating cylinder is represented by state transformation techniques and barrier Lyapunov functions to handle rigid output constraints.⁶ The inertial element of the displacement sensor is governed by:

$$x_d(s) = \frac{K_s}{T_s s + 1} x_p(s)$$

where x_d is the output displacement, K_s is the feedback coefficient, T_s is the time constant of the sensor, and x_p represents the true piston movement.¹²

- **State Variables:** Cylinder stroke, hydraulic fluid pressure, spool valve position.
- **Sensor Variables:** Simulated LVDT position, servo-pressure readouts.
- **Fault Injection:** Simulating internal seal leakage involves introducing a bypass flow coefficient into the state matrix, causing a slow, persistent drift in x_d that forces the control loop into continuous overcompensation.

3. Tandem Mill Kinematics

The rolling force, torque, speed, tension, and thickness reduction are calculated using Orowan's differential equations for flat rolling.²²

- **State Variables:** Strip tension, roll gap, material yield stress, roll speed.
- **Sensor Variables:** Simulated load cell force, motor current, vibration accelerations.
- **Fault Injection:** The generation of the Alpha defect is synthesized by injecting localized, high-density nodes into the simulated steel strip's yield stress matrix, forcing the simulator to output the massive transient force spikes that the ML model must learn to

detect.¹⁷

By fusing these rigorous mathematical simulations with the causal mapping of cross-equipment dependencies, the Digital Twin Input Layer transcends basic anomaly detection. It establishes a fully deterministic, physics-informed synthetic environment capable of generating the exact telemetry signatures of catastrophic failures. This ensures the localized edge AI models are pre-trained to achieve the mandated 100% recall rate, safeguarding the Horizon and Zephyr Hot Rolling Mills against both isolated mechanical breakdowns and cascading, plant-wide production halts.

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